Fig_boxplot

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```r
aggregate_data <- function(data,
                           cols_to_keep,
                           pop_col,
                           agg_noneuca = TRUE){


  pop_cols <- c("pop_ACS", "pop_MPI")

  data_pop <- data %>%
    select(-setdiff(pop_cols, pop_col)) %>%
    rename("pop" = !!sym(pop_col))

  raw_pop <- data_pop %>%
    group_by(aor, region) %>%
    summarize(pop = first(pop), .groups = "drop")

  # find geographic keys
  geo_keys <- intersect(cols_to_keep, c("aor", "region"))

  # aggregate regions if reuqested
  if(agg_noneuca){
    data_pop <- data_pop %>%
      mutate(region = ifelse(region == "EUCA", "EUCA", "nonEUCA")) %>%
      group_by(pick(-c(n_arrests, pop))) %>%
      summarize(n_arrests = sum(n_arrests),
                pop = sum(pop))

    raw_pop <- raw_pop %>%
      mutate(region = ifelse(region == "EUCA", "EUCA", "nonEUCA")) %>%
      group_by(aor, region) %>%
      summarize(pop = sum(pop), .groups = "drop")
  }

  # population map
  pop_map <- data_pop %>%
    group_by(aor, region, pop) %>%
    slice_head(n = 1) %>%
    ungroup() %>%
    group_by_at(geo_keys) %>%
    summarize(pop = sum(pop), .groups = "drop")

  # checksum value
```

```r
  checksum <- data_pop %>%
    group_by(aor, region, pop) %>%
    slice_head(n=1) %>%
    ungroup() %>%
    group_by(region) %>%
    summarize(pop_checksum = sum(pop), .groups = "drop")

  # aggregate arrests, join population back in
  data_pop <- data_pop %>%
    group_by_at(cols_to_keep) %>%
    summarize(n_arrests = sum(n_arrests, na.rm = TRUE), .groups = "drop") %>%
    left_join(pop_map, by = geo_keys) %>%
    mutate(ratio = n_arrests/pop*1000,
           log_pop = log(pop))

  # checksum
  total_aggregated <- data_pop %>%
    group_by_at(geo_keys) %>%
    summarize(pop = first(pop), .groups = "drop") %>%
    pull(pop) %>%
    sum(na.rm = TRUE)
  total_checksum <- sum(checksum$pop_checksum, na.rm = TRUE)
  if (!(dplyr::near(total_checksum, total_aggregated))) {
    warning("Checksum Failed!")
  }

  return(data_pop)
}
```

```r
trajectory <- read_csv("../data/processed/glm_trajectory_type_threatlevel_convicted_ACS_MPI_AOR.csv")
```

```
## Rows: 113400 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (4): aor, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

```r
trajectory %>%
  filter(region != "EUCA") %>%
  ungroup() %>%
  filter(window == 0 & threat_level == 0 & convicted == 0 & method == "CA" & alt_method == "LEA") %>%
  summarize(sum(pop_MPI))
```

```
## # A tibble: 1 x 1
##   `sum(pop_MPI)`
##            <dbl>
## 1       12829000
```

```r
conv_method_aor <- aggregate_data(trajectory, c("region", "aor", "window"), "pop_MPI", TRUE) %>%
  mutate(window = ifelse(window < 0, -1, 0)) %>%
  group_by(window, region, aor, pop) %>%
  summarize(n_arrests = sum(n_arrests)) %>%
```

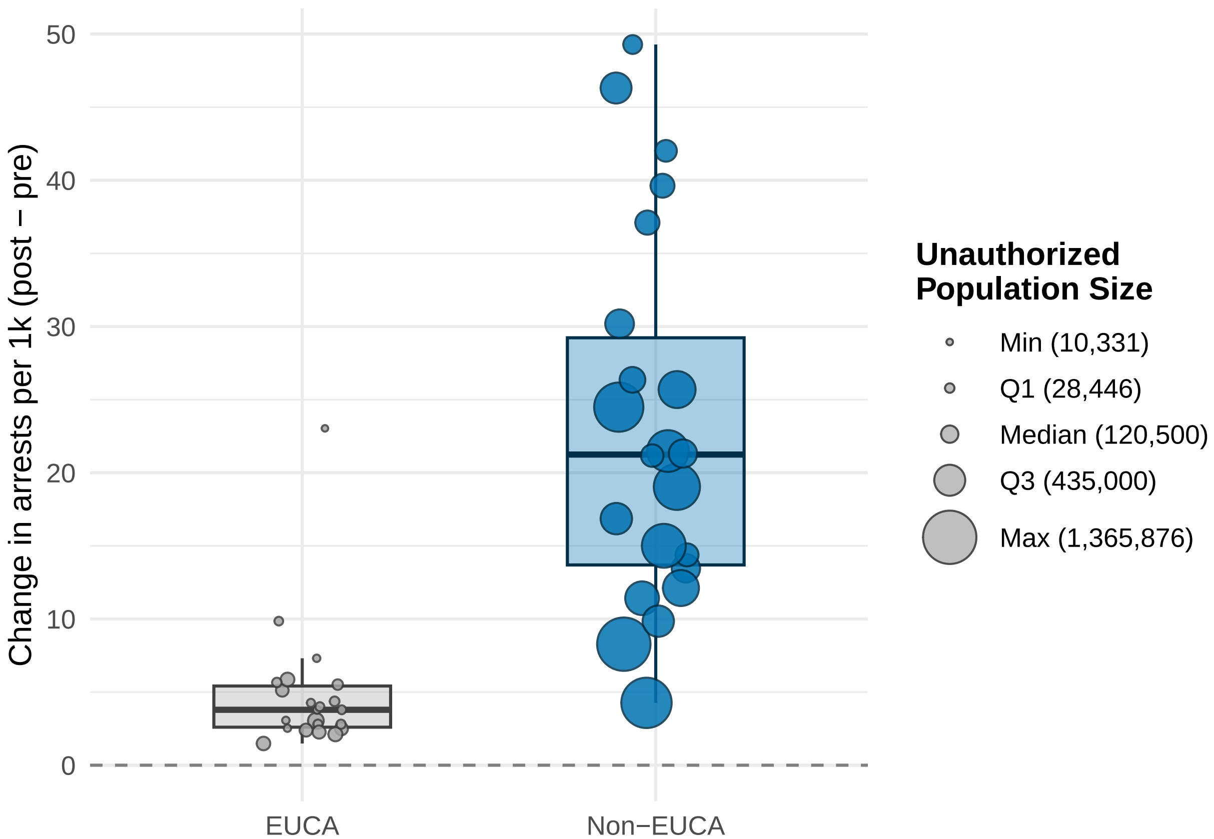
```r
  mutate(ratio = n_arrests/pop *1000)
```

```
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by aor, region, window, convicted,
##   threat_level, method, and alt_method.
## i Output is grouped by aor, region, window, convicted, threat_level, and
##   method.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(aor, region, window, convicted, threat_level, method,
##   alt_method))` for per-operation grouping (`?dplyr::dplyr_by`) instead.
```

```r
aors_above_threshold_euca <- (conv_method_aor %>%
  filter(region == "EUCA" & window == "-1" & pop > 10000)) %>%
  ungroup() %>%
  select(aor, pop)

aors_above_threshold_noneuca <- (conv_method_aor %>%
  filter(region == "nonEUCA" & window == "-1" & aor %in% aors_above_threshold_euca$aor)) %>%
  ungroup() %>%
  select(aor, pop)

conv_method_aor_euca <- conv_method_aor %>%
  filter(region == "EUCA" & aor %in% aors_above_threshold_euca$aor) %>%
  pivot_wider(id_cols = aor, names_from = window, values_from = ratio) %>%
  mutate(diff = `0`-`-1`) %>%
  mutate(region = "EUCA") %>%
  merge(aors_above_threshold_euca)

conv_method_aor_noneuca <- conv_method_aor %>%
  filter(region == "nonEUCA" & aor %in% aors_above_threshold_noneuca$aor) %>%
  pivot_wider(id_cols = aor, names_from = window, values_from = ratio) %>%
  mutate(diff = `0`-`-1`) %>%
  mutate(region = "Non-EUCA") %>%
  merge(aors_above_threshold_noneuca)

conv_method <- rbind(conv_method_aor_euca, conv_method_aor_noneuca)

gaps_widen <- conv_method_aor_euca %>%
  select(aor, diff) %>%
  rename("diff_euca" = "diff") %>%
  merge(conv_method_aor_noneuca %>% select(aor, diff)) %>%
  mutate(diff = diff-diff_euca)
```

```r
jpos <- position_jitter(width = 0.12, height = 0, seed = 78483)


pop_breaks <- quantile(conv_method$pop, probs = c(0, 0.25, 0.5, 0.75, 1), na.rm = TRUE)
pop_labels <- c("Min", "Q1", "Median", "Q3", "Max")

pop_labels <- paste0(
  c("Min", "Q1", "Median", "Q3", "Max"),
  " (", comma(pop_breaks), ")"
)
```

```r
ggplot(conv_method, aes(x = region, y = diff)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey50") +
  geom_boxplot(aes(fill = region, color = region), width = 0.5, alpha = 0.35, outlier.shape = NA) +
  geom_jitter(aes(size = pop, fill = region, color = region), position = jpos, alpha = 0.85, shape = 21
  scale_size_area(
    name = "Unauthorized\nPopulation Size",
    breaks = pop_breaks,
    labels = paste0(pop_labels),
    max_size = 9,
    guide = guide_legend(
      override.aes = list(
        fill = "grey75",
        color = "grey30",
        alpha = 1
      )
    )
  ) +
  scale_fill_manual(values = c("EUCA" = "grey65", "Non-EUCA" = "#0072B2"), guide = "none") +
  scale_color_manual(values = c("EUCA" = "grey25", "Non-EUCA" = "#003049"), guide = "none") +
  labs(x = NULL, y = "Change in arrests per 1k (post - pre)") +
  theme_minimal(base_size = 12) +
  theme(
    legend.position = "right",
    legend.title = element_text(face = "bold")
  )
```



```r
ggsave("../figures/main_text/mpi_denoms_fig3.pdf", height = 4.5, width = 7.5)
```

Fetch IQRs

```r
quantile(conv_method[conv_method$region == "EUCA",]$diff , probs = c(.25, .75))
```

```
##      25%      75%
## 2.600089 5.415346
```

```r
quantile(conv_method[conv_method$region == "Non-EUCA",]$diff , probs = c(.25, .75))
```

```
##      25%      75%
## 13.69571 29.22980
```

```r
trajectory <- read_csv("../data/processed/glm_trajectory_type_threatlevel_convicted_ACS_MPI_AOR.csv")
```

```
## Rows: 113400 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (4): aor, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

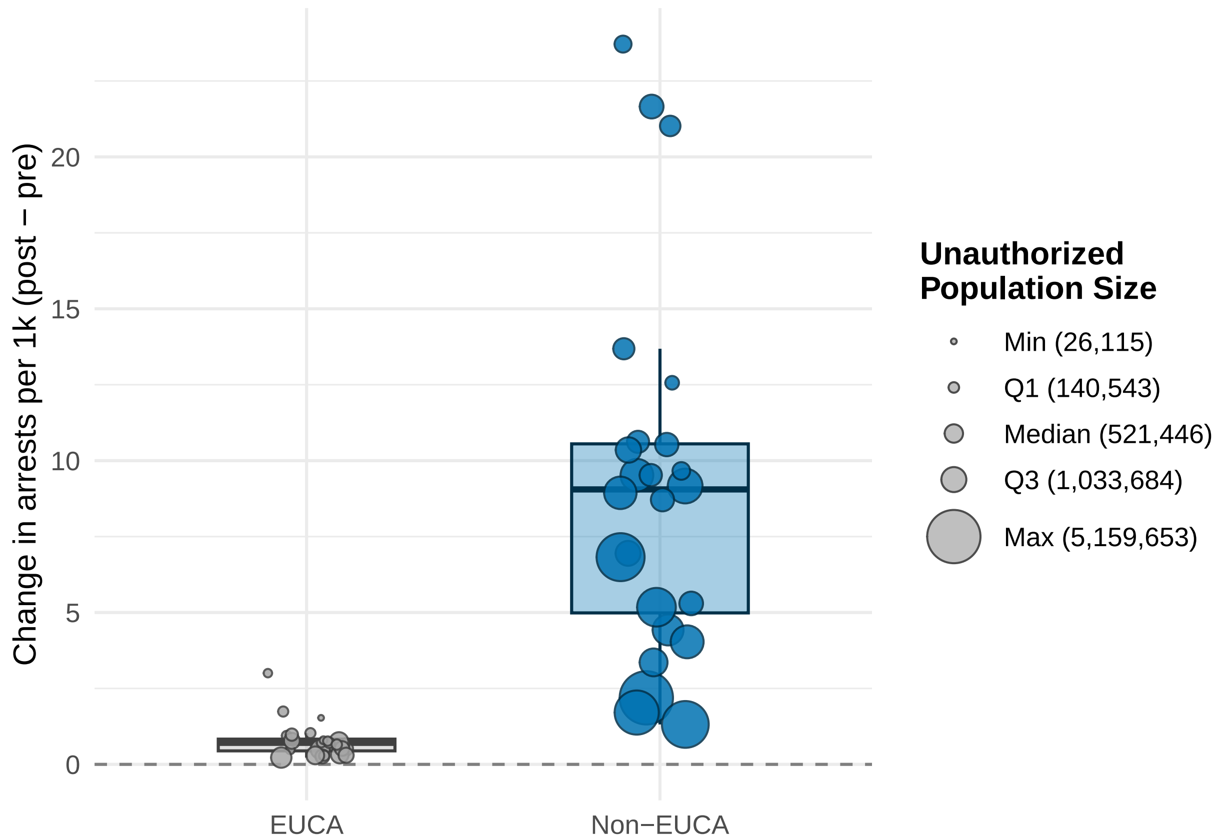
```r
conv_method_aor <- aggregate_data(trajectory, c("region", "aor", "window"), "pop_ACS", TRUE) %>%
  mutate(window = ifelse(window < 0, -1, 0)) %>%
  group_by(window, region, aor, pop) %>%
  summarize(n_arrests = sum(n_arrests)) %>%
  mutate(ratio = n_arrests/pop *1000)
```

```
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by aor, region, window, convicted,
##   threat_level, method, and alt_method.
## i Output is grouped by aor, region, window, convicted, threat_level, and
##   method.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(aor, region, window, convicted, threat_level, method,
##   alt_method))` for per-operation grouping (`?dplyr::dplyr_by`) instead.
```

```r
aors_above_threshold_euca <- (conv_method_aor %>%
  filter(region == "EUCA" & window == "-1" & pop > 10000)) %>%
  ungroup() %>%
  select(aor, pop)

aors_above_threshold_noneuca <- (conv_method_aor %>%
  filter(region == "nonEUCA" & window == "-1" & aor %in% aors_above_threshold_euca$aor)) %>%
  ungroup() %>%
  select(aor, pop)

conv_method_aor_euca <- conv_method_aor %>%
  filter(region == "EUCA" & aor %in% aors_above_threshold_euca$aor) %>%
  pivot_wider(id_cols = aor, names_from = window, values_from = ratio) %>%
  mutate(diff = `0`-`-1`) %>%
  mutate(region = "EUCA") %>%
  merge(aors_above_threshold_euca)
```

```r
conv_method_aor_noneuca <- conv_method_aor %>%
  filter(region == "nonEUCA" & aor %in% aors_above_threshold_noneuca$aor) %>%
  pivot_wider(id_cols = aor, names_from = window, values_from = ratio) %>%
  mutate(diff = `0`-`-1`) %>%
  mutate(region = "Non-EUCA") %>%
  merge(aors_above_threshold_noneuca)

conv_method <- rbind(conv_method_aor_euca, conv_method_aor_noneuca)

gaps_widen <- conv_method_aor_euca %>%
  select(aor, diff) %>%
  rename("diff_euca" = "diff") %>%
  merge(conv_method_aor_noneuca %>% select(aor, diff)) %>%
  mutate(diff = diff-diff_euca)
```

```r
jpos <- position_jitter(width = 0.12, height = 0, seed = 78483)


pop_breaks <- quantile(conv_method$pop, probs = c(0, 0.25, 0.5, 0.75, 1), na.rm = TRUE)
pop_labels <- c("Min", "Q1", "Median", "Q3", "Max")

pop_labels <- paste0(
  c("Min", "Q1", "Median", "Q3", "Max"),
  " (", comma(pop_breaks), ")"
)

ggplot(conv_method, aes(x = region, y = diff)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey50") +
  geom_boxplot(aes(fill = region, color = region), width = 0.5, alpha = 0.35, outlier.shape = NA) +
  geom_jitter(aes(size = pop, fill = region, color = region), position = jpos, alpha = 0.85, shape = 21
  scale_size_area(
    name = "Unauthorized\nPopulation Size",
    breaks = pop_breaks,
    labels = paste0(pop_labels),
    max_size = 9,
    guide = guide_legend(
      override.aes = list(
        fill = "grey75",
        color = "grey30",
        alpha = 1
      )
    )
  ) +
  scale_fill_manual(values = c("EUCA" = "grey65", "Non-EUCA" = "#0072B2"), guide = "none") +
  scale_color_manual(values = c("EUCA" = "grey25", "Non-EUCA" = "#003049"), guide = "none") +
  labs(x = NULL, y = "Change in arrests per 1k (post - pre)") +
  theme_minimal(base_size = 12) +
  theme(
    legend.position = "right",
    legend.title = element_text(face = "bold")
  )
```




```r
ggsave("../figures/supplement/acs_denoms_fig3.pdf", height = 4.5, width = 7.5)
```

```r
trajectory <- read_csv("../data/processed/glm_trajectory_type_threatlevel_convicted_ACS_MPI_AOR.csv")
```

```
## Rows: 113400 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (4): aor, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

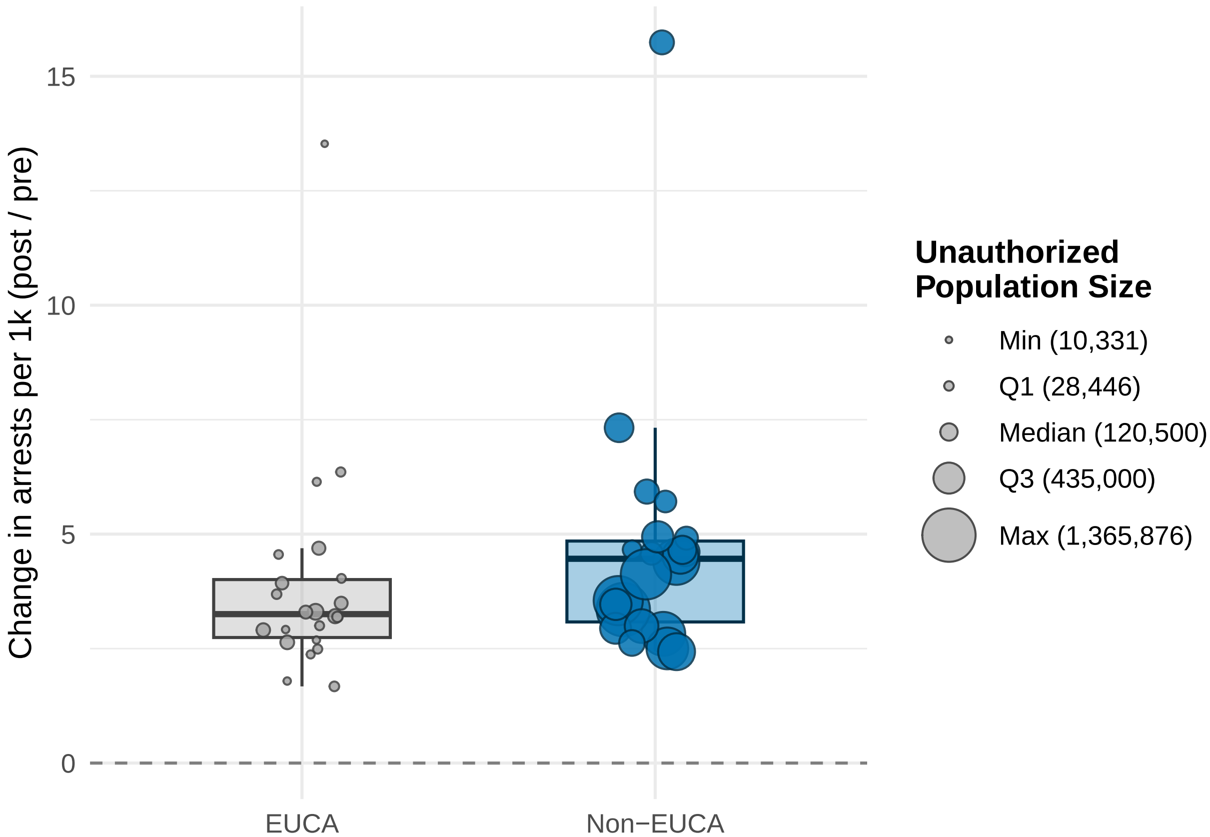
```r
conv_method_aor <- aggregate_data(trajectory, c("region", "aor", "window"), "pop_MPI", TRUE) %>%
  mutate(window = ifelse(window < 0, -1, 0)) %>%
  group_by(window, region, aor, pop) %>%
  summarize(n_arrests = sum(n_arrests)) %>%
  mutate(ratio = n_arrests/pop *1000)
```

```
## `summarise()` has regrouped the output.
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by aor, region, window, convicted,
##   threat_level, method, and alt_method.
## i Output is grouped by aor, region, window, convicted, threat_level, and
##   method.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(aor, region, window, convicted, threat_level, method,
##   alt_method))` for per-operation grouping (`?dplyr::dplyr_by`) instead.
```

```r
aors_above_threshold_euca <- (conv_method_aor %>%
  filter(region == "EUCA" & window == "-1" & pop > 10000)) %>%
  ungroup() %>%
  select(aor, pop)

aors_above_threshold_noneuca <- (conv_method_aor %>%
  filter(region == "nonEUCA" & window == "-1" & aor %in% aors_above_threshold_euca$aor)) %>%
  ungroup() %>%
  select(aor, pop)

conv_method_aor_euca <- conv_method_aor %>%
  filter(region == "EUCA" & aor %in% aors_above_threshold_euca$aor) %>%
  pivot_wider(id_cols = aor, names_from = window, values_from = ratio) %>%
  mutate(diff = `0`/`-1`) %>%
  mutate(region = "EUCA") %>%
  merge(aors_above_threshold_euca)

conv_method_aor_noneuca <- conv_method_aor %>%
  filter(region == "nonEUCA" & aor %in% aors_above_threshold_noneuca$aor) %>%
  pivot_wider(id_cols = aor, names_from = window, values_from = ratio) %>%
  mutate(diff = `0`/`-1`) %>%
  mutate(region = "Non-EUCA") %>%
  merge(aors_above_threshold_noneuca)

conv_method <- rbind(conv_method_aor_euca, conv_method_aor_noneuca)

gaps_widen <- conv_method_aor_euca %>%
  select(aor, diff) %>%
  rename("diff_euca" = "diff") %>%
  merge(conv_method_aor_noneuca %>% select(aor, diff)) %>%
  mutate(diff = diff/diff_euca)
```

```r
jpos <- position_jitter(width = 0.12, height = 0, seed = 78483)


pop_breaks <- quantile(conv_method$pop, probs = c(0, 0.25, 0.5, 0.75, 1), na.rm = TRUE)
pop_labels <- c("Min", "Q1", "Median", "Q3", "Max")

pop_labels <- paste0(
  c("Min", "Q1", "Median", "Q3", "Max"),
  " (", comma(pop_breaks), ")"
)

ggplot(conv_method, aes(x = region, y = diff)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey50") +
  geom_boxplot(aes(fill = region, color = region), width = 0.5, alpha = 0.35, outlier.shape = NA) +
  geom_jitter(aes(size = pop, fill = region, color = region), position = jpos, alpha = 0.85, shape = 21
  scale_size_area(
    name = "Unauthorized\nPopulation Size",
    breaks = pop_breaks,
    labels = paste0(pop_labels),
    max_size = 9,
    guide = guide_legend(
      override.aes = list(
```

```r
        fill = "grey75",
        color = "grey30",
        alpha = 1
      )
    )
  ) +
  scale_fill_manual(values = c("EUCA" = "grey65", "Non-EUCA" = "#0072B2"), guide = "none") +
  scale_color_manual(values = c("EUCA" = "grey25", "Non-EUCA" = "#003049"), guide = "none") +
  labs(x = NULL, y = "Change in arrests per 1k (post / pre)") +
  theme_minimal(base_size = 12) +
  theme(
    legend.position = "right",
    legend.title = element_text(face = "bold")
  )
```




```r
ggsave("../figures/supplement/multiplicative_fig3.pdf", height = 4.5, width = 7.5)
```

```r
aggregate_data <- function(data,
                           cols_to_keep,
                           pop_col,
                           agg_noneuca = TRUE){


  pop_cols <- c("pop_ACS", "pop_MPI")

  data_pop <- data %>%
    select(-setdiff(pop_cols, pop_col)) %>%
```

```r
  rename("pop" = !!sym(pop_col))

raw_pop <- data_pop %>%
  group_by(state, region) %>%
  summarize(pop = first(pop), .groups = "drop")

# aggregate regions if reuqested
if(agg_noneuca){
  data_pop <- data_pop %>%
    mutate(region = ifelse(region == "EUCA", "EUCA", "nonEUCA"))

  raw_pop <- raw_pop %>%
    mutate(region = ifelse(region == "EUCA", "EUCA", "nonEUCA")) %>%
    group_by(state, region) %>%
    summarize(pop = sum(pop), .groups = "drop")
}

# checksum value
checksum <- data_pop %>%
  group_by(state, region, pop) %>%
  slice_head(n=1) %>%
  ungroup() %>%
  group_by(region) %>%
  summarize(pop_checksum = sum(pop), .groups = "drop")

# find geographic keys
geo_keys <- intersect(cols_to_keep, c("state", "region"))

# population map
pop_map <- data_pop %>%
  group_by(state, region, pop) %>%
  slice_head(n = 1) %>%
  ungroup() %>%
  group_by_at(geo_keys) %>%
  summarize(pop = sum(pop), .groups = "drop")

# aggregate arrests, join population back in
data_pop <- data_pop %>%
  group_by_at(cols_to_keep) %>%
  summarize(n_arrests = sum(n_arrests, na.rm = TRUE), .groups = "drop") %>%
  left_join(pop_map, by = geo_keys) %>%
  mutate(ratio = n_arrests/pop*1000,
         log_pop = log(pop))

# checksum
total_aggregated <- data_pop %>%
  group_by_at(geo_keys) %>%
  summarize(pop = first(pop), .groups = "drop") %>%
  pull(pop) %>%
  sum(na.rm = TRUE)
total_checksum <- sum(checksum$pop_checksum, na.rm = TRUE)
if (!(dplyr::near(total_checksum, total_aggregated))) {
  warning("Checksum Failed!")
```

```r
  }

  return(data_pop)
}
```

```r
trajectory <- read_csv("../data/processed/glm_trajectory_type_threatlevel_convicted_ACS_MPI.csv")
```

```
## Rows: 231336 Columns: 10
## -- Column specification --------------------------------------------------------
## Delimiter: ","
## chr (4): state, region, method, alt_method
## dbl (6): window, convicted, threat_level, n_arrests, pop_ACS, pop_MPI
##
## i Use `spec()` to retrieve the full column specification for this data.
## i Specify the column types or set `show_col_types = FALSE` to quiet this message.
```

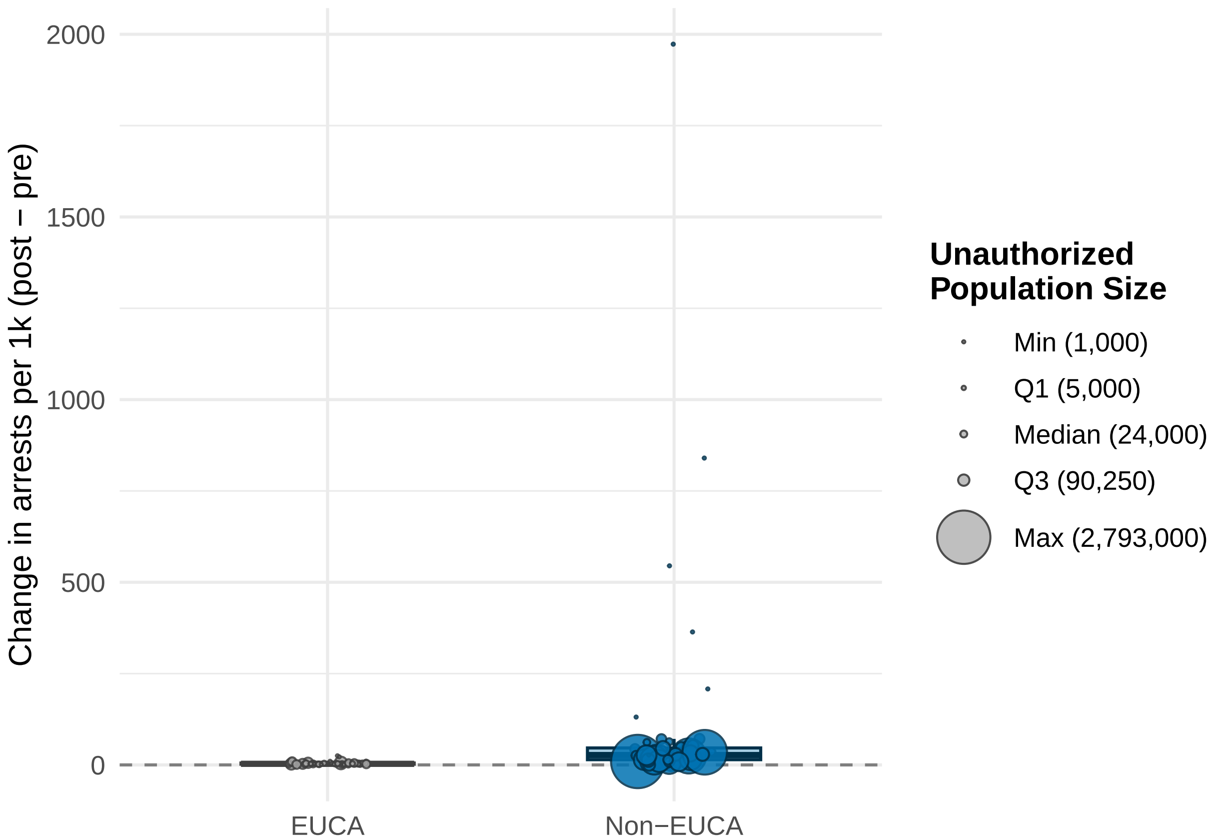
```r
conv_method_state <- aggregate_data(trajectory, c("region", "state", "window"), "pop_MPI", TRUE) %>%
  mutate(window = ifelse(window < 0, -1, 0)) %>%
  group_by(window, region, state, pop) %>%
  summarize(n_arrests = sum(n_arrests)) %>%
  mutate(ratio = n_arrests/pop *1000)
```

```
## `summarise()` has regrouped the output.
## i Summaries were computed grouped by window, region, state, and pop.
## i Output is grouped by window, region, and state.
## i Use `summarise(.groups = "drop_last")` to silence this message.
## i Use `summarise(.by = c(window, region, state, pop))` for per-operation
##   grouping (`?dplyr::dplyr_by`) instead.
```

```r
states_above_threshold_euca <- (conv_method_state %>%
  filter(region == "EUCA" & window == "-1" & pop > 0)) %>%
  ungroup() %>%
  select(state, pop)

states_above_threshold_noneuca <- (conv_method_state %>%
  filter(region == "nonEUCA" & window == "-1" & state %in% states_above_threshold_euca$state)) %>%
  ungroup() %>%
  select(state, pop)

conv_method_state_euca <- conv_method_state %>%
  filter(region == "EUCA" & state %in% states_above_threshold_euca$state) %>%
  pivot_wider(id_cols = state, names_from = window, values_from = ratio) %>%
  mutate(diff = `0`-`-1`) %>%
  mutate(region = "EUCA") %>%
  merge(states_above_threshold_euca)

conv_method_state_noneuca <- conv_method_state %>%
  filter(region == "nonEUCA" & state %in% states_above_threshold_noneuca$state) %>%
  pivot_wider(id_cols = state, names_from = window, values_from = ratio) %>%
  mutate(diff = `0`-`-1`) %>%
  mutate(region = "Non-EUCA") %>%
  merge(states_above_threshold_noneuca)

conv_method <- rbind(conv_method_state_euca, conv_method_state_noneuca)
```

```
gaps_widen <- conv_method_state_euca %>%
  select(state, diff) %>%
  rename("diff_euca" = "diff") %>%
  merge(conv_method_state_noneuca %>% select(state, diff)) %>%
  mutate(diff = diff-diff_euca)
```

```
jpos <- position_jitter(width = 0.12, height = 0, seed = 78483)


pop_breaks <- quantile(conv_method$pop, probs = c(0, 0.25, 0.5, 0.75, 1), na.rm = TRUE)
pop_labels <- c("Min", "Q1", "Median", "Q3", "Max")

pop_labels <- paste0(
  c("Min", "Q1", "Median", "Q3", "Max"),
  " (", comma(pop_breaks), ")"
)

ggplot(conv_method, aes(x = region, y = diff)) +
  geom_hline(yintercept = 0, linetype = "dashed", color = "grey50") +
  geom_boxplot(aes(fill = region, color = region), width = 0.5, alpha = 0.35, outlier.shape = NA) +
  geom_jitter(aes(size = pop, fill = region, color = region), position = jpos, alpha = 0.85, shape = 21)
  scale_size_area(
    name = "Unauthorized\nPopulation Size",
    breaks = pop_breaks,
    labels = paste0(pop_labels),
    max_size = 9,
    guide = guide_legend(
      override.aes = list(
        fill = "grey75",
        color = "grey30",
        alpha = 1
      )
    )
  ) +
  scale_fill_manual(values = c("EUCA" = "grey65", "Non-EUCA" = "#0072B2"), guide = "none") +
  scale_color_manual(values = c("EUCA" = "grey25", "Non-EUCA" = "#003049"), guide = "none") +
  labs(x = NULL, y = "Change in arrests per 1k (post - pre)") +
  theme_minimal(base_size = 12) +
  theme(
    legend.position = "right",
    legend.title = element_text(face = "bold")
  )
```


Change in arrests per 1k (post − pre)
2000
1500
1000
500
0
EUCA
Non−EUCA
Unauthorized Population Size
Min (1,000)
Q1 (5,000)
Median (24,000)
Q3 (90,250)
Max (2,793,000)